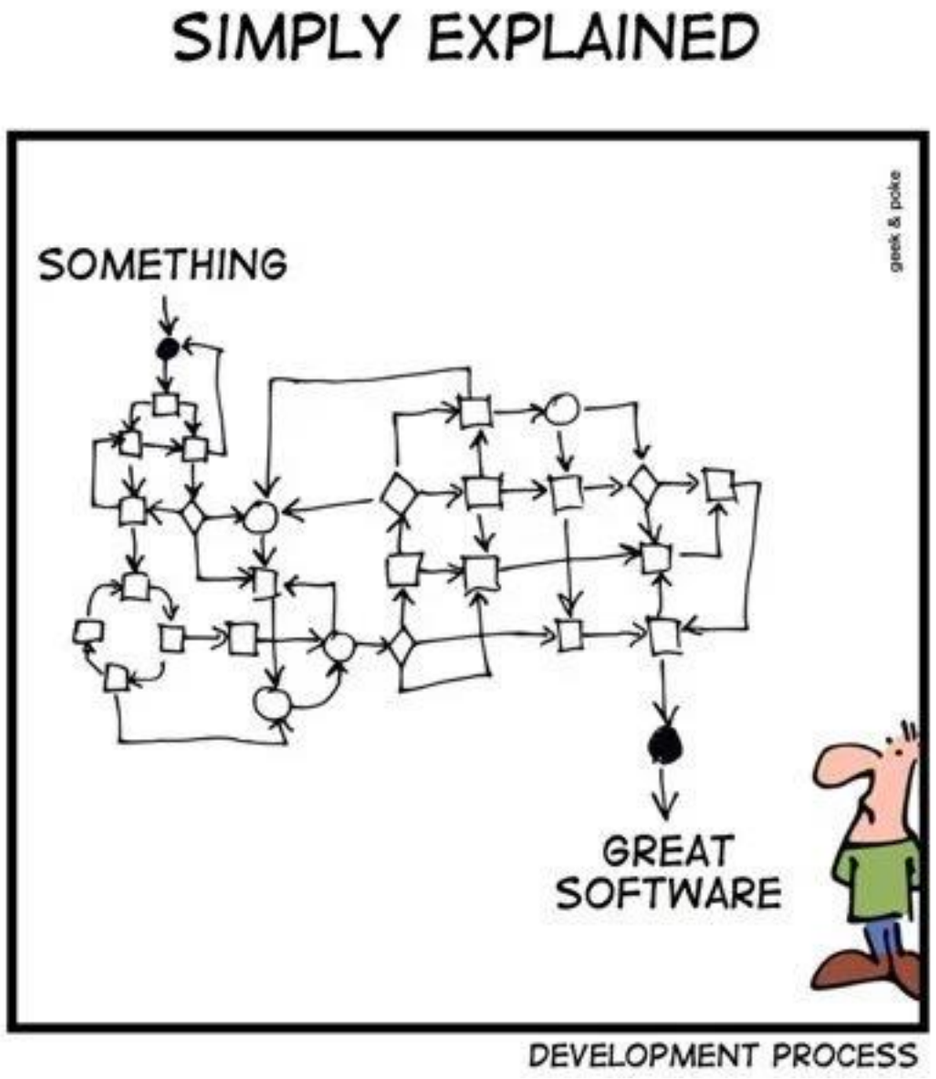
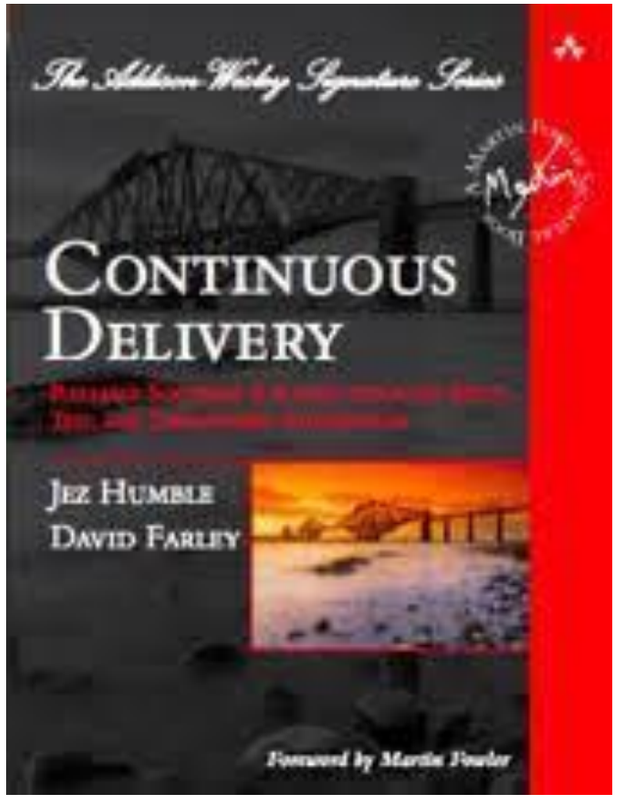
Developer Operations (devops)

Building Software

Building Software



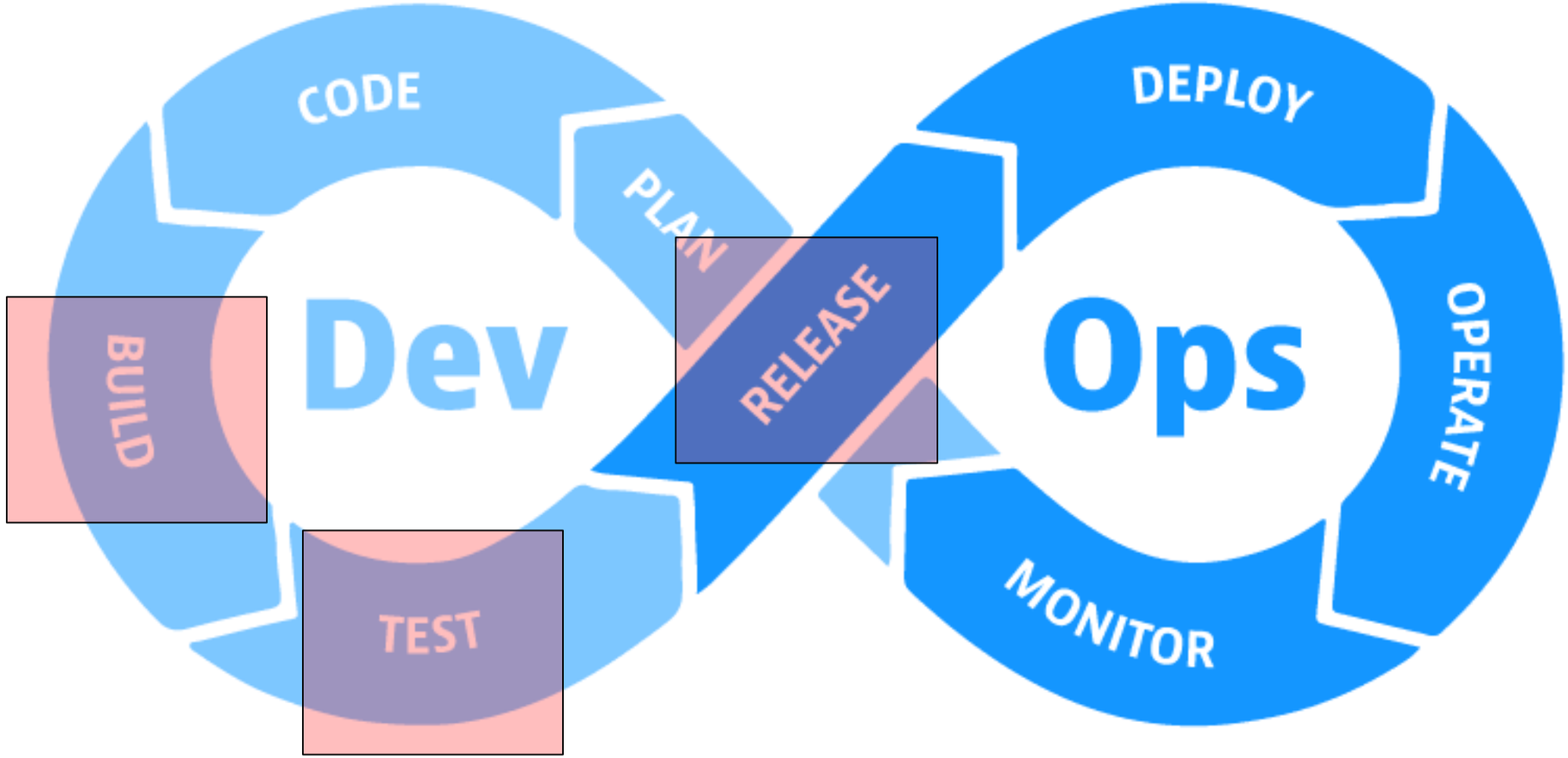
Building Software, Principles of Software Delivery



1. ***Create a Repeatable, Reliable Process for Releasing Software***
2. ***Automate Almost Everything***
3. ***Keep Everything in Version Control***
4. ***If It Hurts, Do It More Frequently, and Bring the Pain Forward***
5. ***Build Quality In***
6. ***Done Means Released***
7. ***Everybody Is Responsible for the Delivery Process***
8. ***Continuous Improvement***

We will use Gitlab CI for creating a suitable release process satisfying these points.

Build and Release



What tests do you want to continuously run?

What analysis do you want to continuously perform?

What is a release?

Release != Deployments

Separation of Concerns:

– Different Domains / Tools for generating artifact and deploying them

Quick, painless builds:

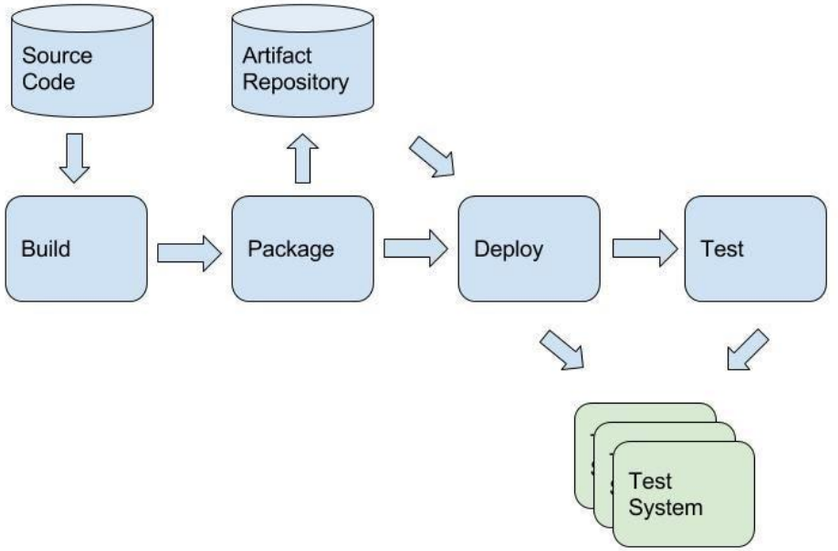
– Builds (and thereby releases) should be automated and painless just generating artifacts

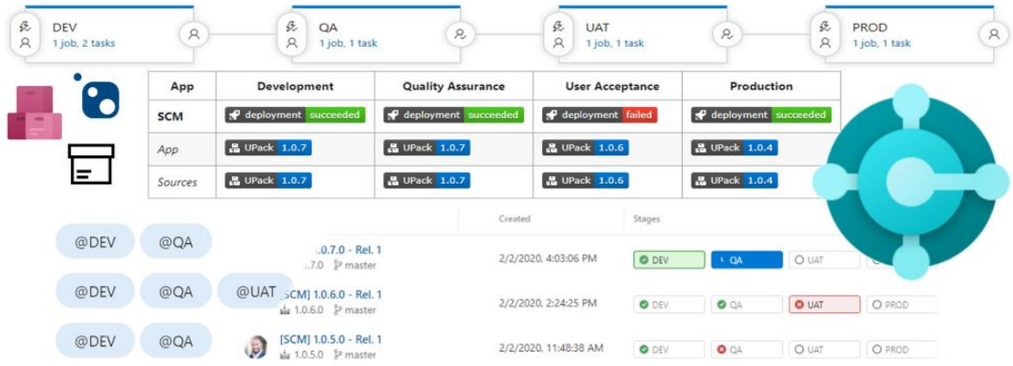
Independent deployable and immutable artifacts:

– Deployment Units should be independent from environment (stage)

Idempotent and scalable tests:

– By separating deployment from releasing, scalable tests on the same artifact become possible





What is a release in frameworks?

Maven

- Identified by fixed version in GAV(C)
- Unique with respect to GAV(C)
- Not to be modified again
  - Seen as “stable”
  - Proctected for overriding artifact

Otherwise, transitive dependencies will break

For continuous development,
please use “-SNAPSHOT”-suffix

Docker / OCI

- Identified by Tags
- Unique with respect of NAME:TAG
- Can be overridden, seldomly overridden

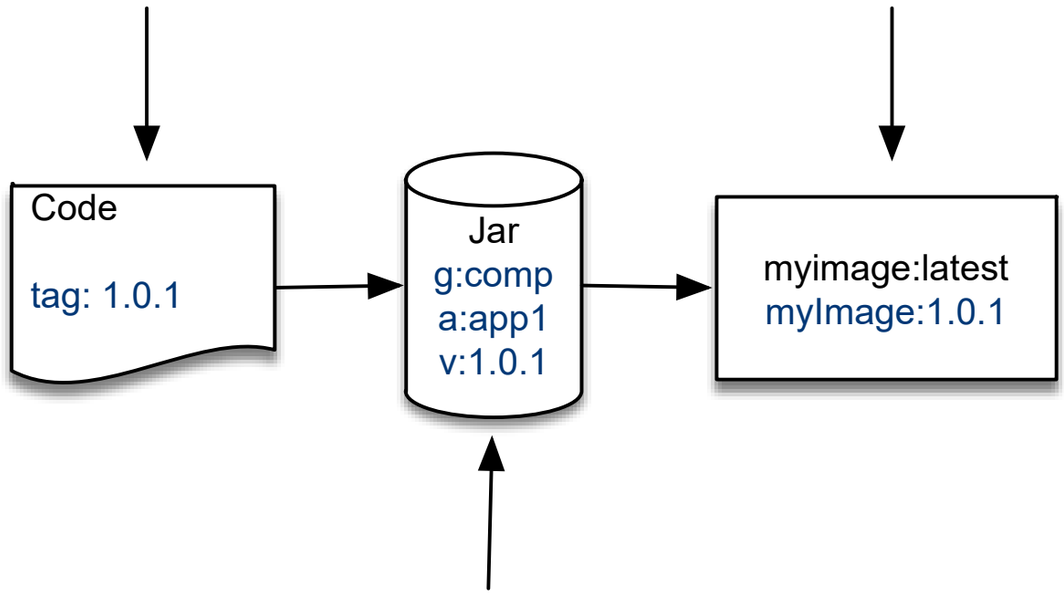
- No danger of breaking transitive dependencies -> Images are self-contained

For continuous development,
please use the tag “latest”

Releases can be complex…

```
>git tag 1.0.1
>git push —tags
```

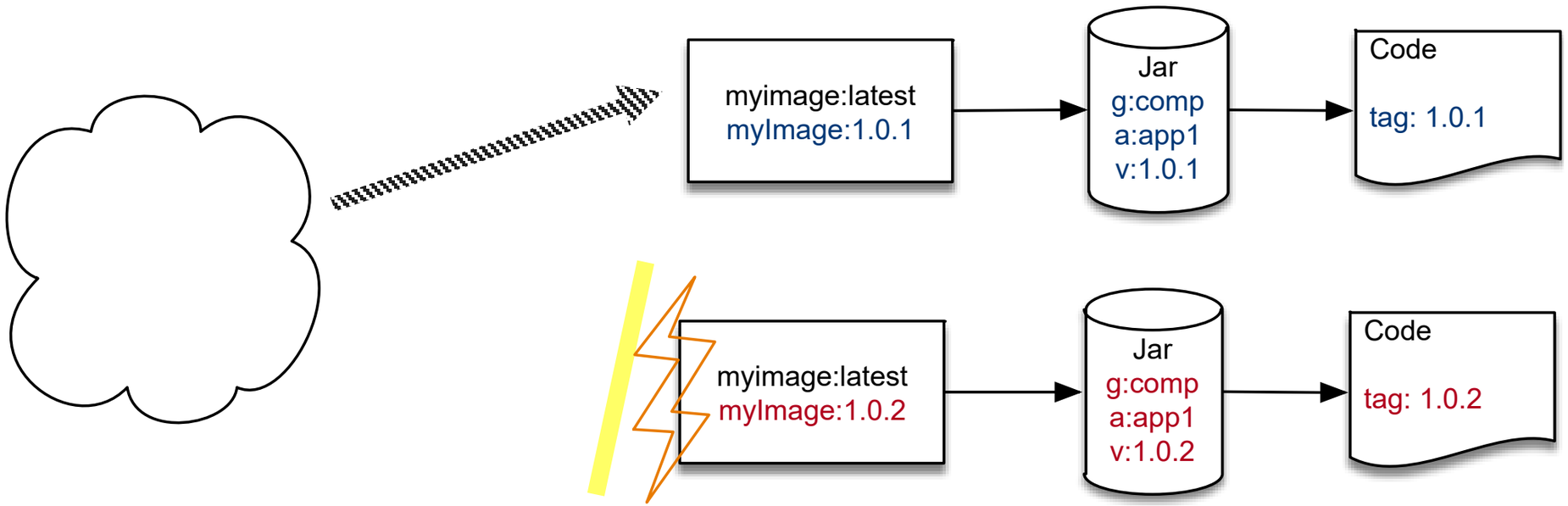
```
>docker tag myImage:1.0.1
>docker push myImage:1.0.1
```



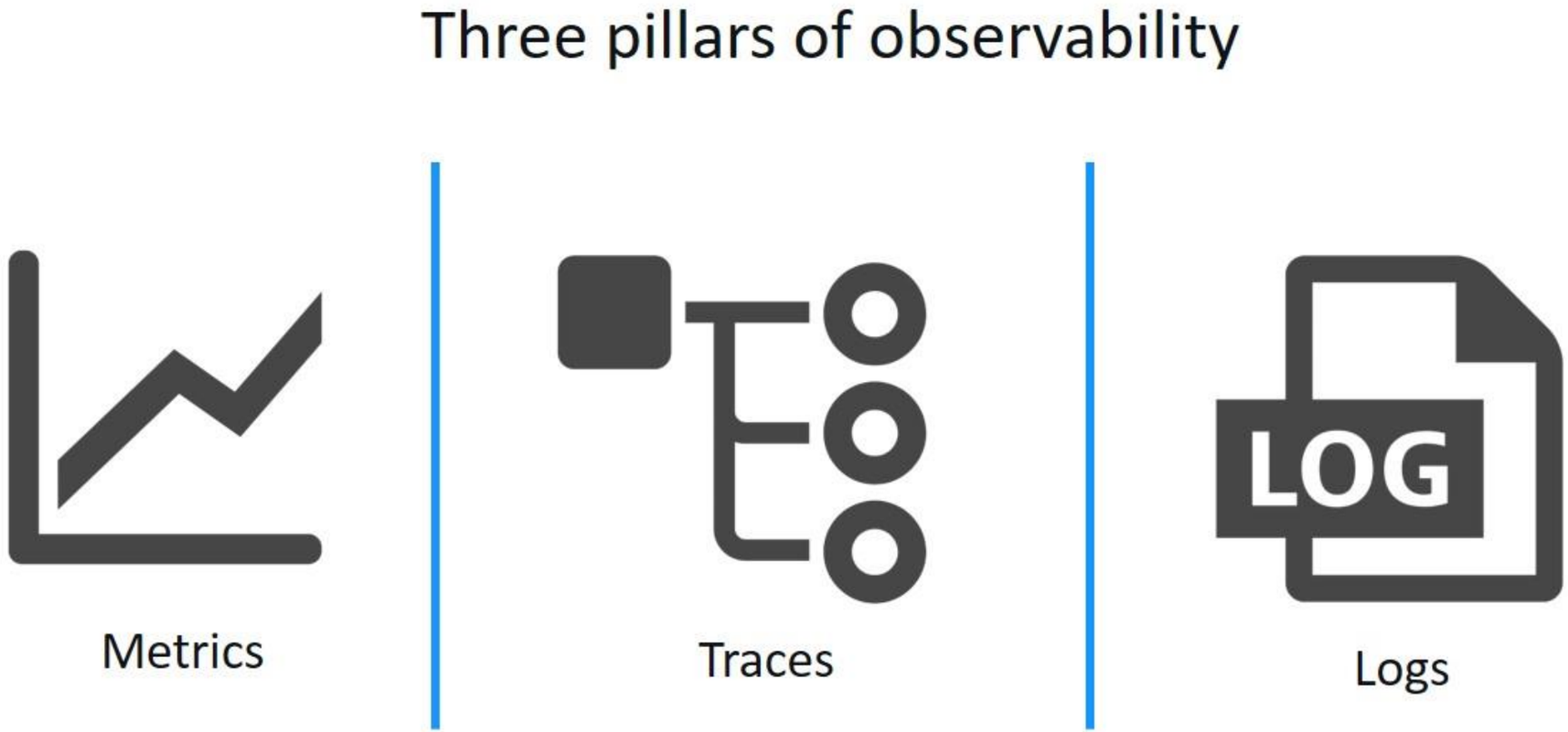

```
>mvn release:perform -DreleaseVersion=1.0.1
```

What is a problem when releasing on a CI?

Why is regular releasing important?



Why?



Repeadible Builds

or “what is the problem with maven/pip/go mod”

Build Systems are programming language-centric:

- How to compile to get an executable?
- How to handle dependencies?

However:

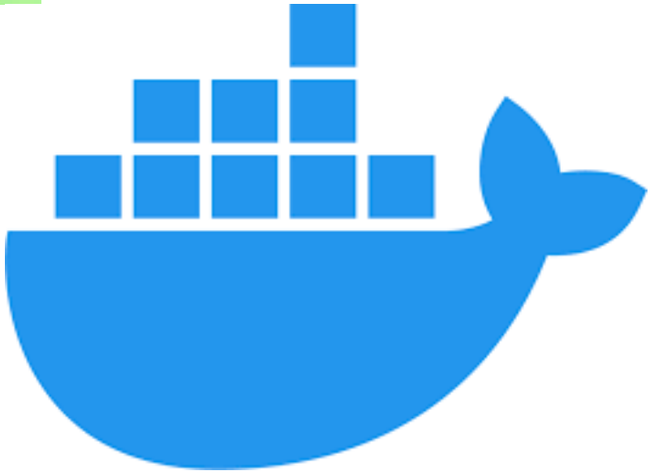
- Common knowledge about the system is needed
- Each System behaves differently

Repeatable Builds with the help of an Independent Build System

Builds all systems the same way:

- Clear toolset leveraging from native build systems
- Easy integration into CI/CD-systems
- Common knowledge about NFAs of build systems:
  - Caching
  - Syntax
  - Workflow
  - …

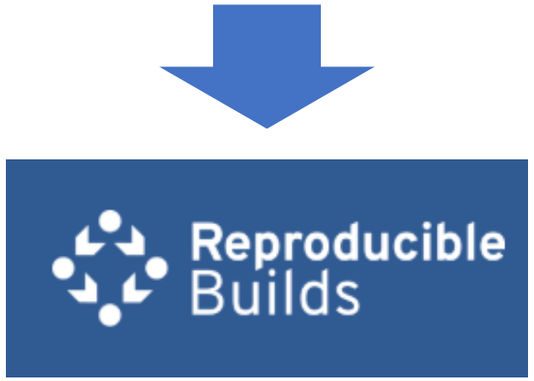
Dagger

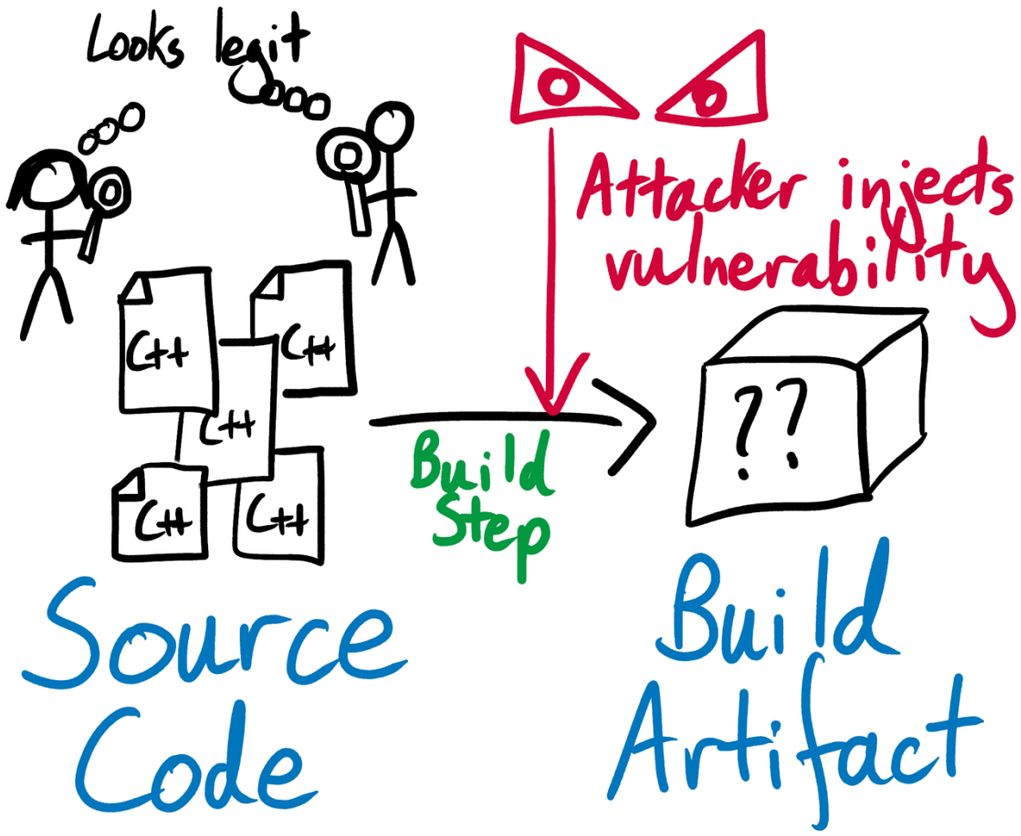


Reproducible Builds

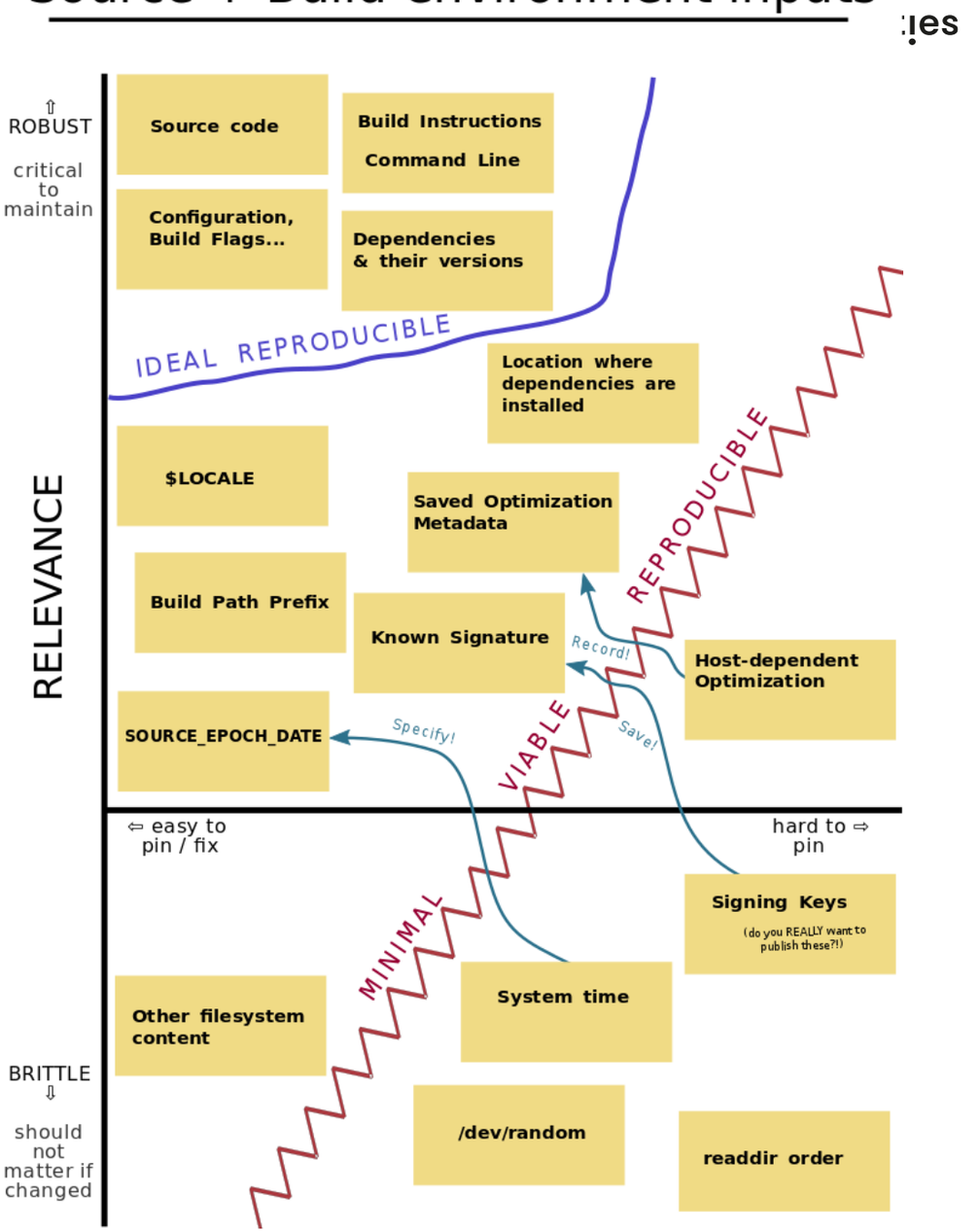
https://reproducible-builds.org/

"The build system should be able to take the build inputs (source code, assets, and so on) and produce repeatable artifacts. The code built from the same inputs last week should produce the same output this week"
– Google SRE Workbook

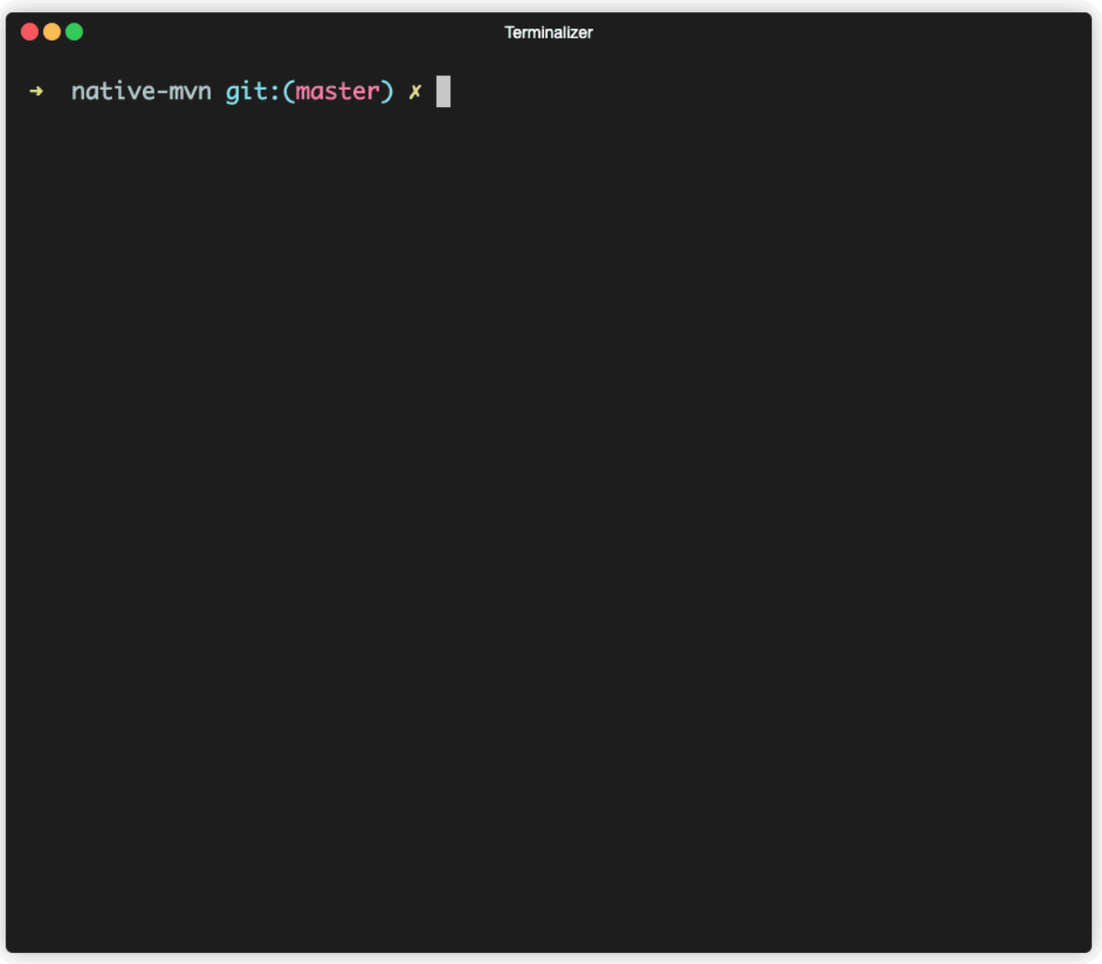


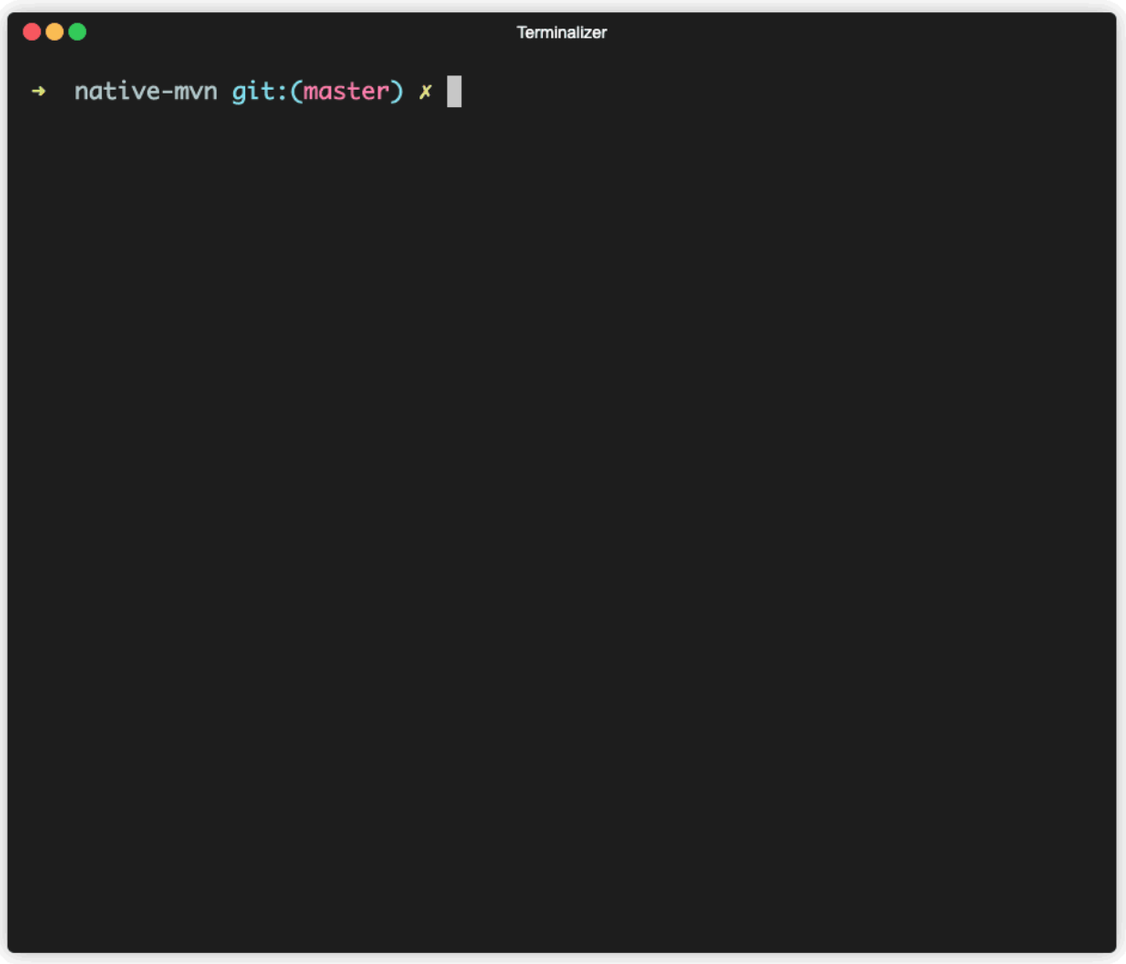


Reproducible Builds hard to achieve



Example: Maven and Reproducible Builds



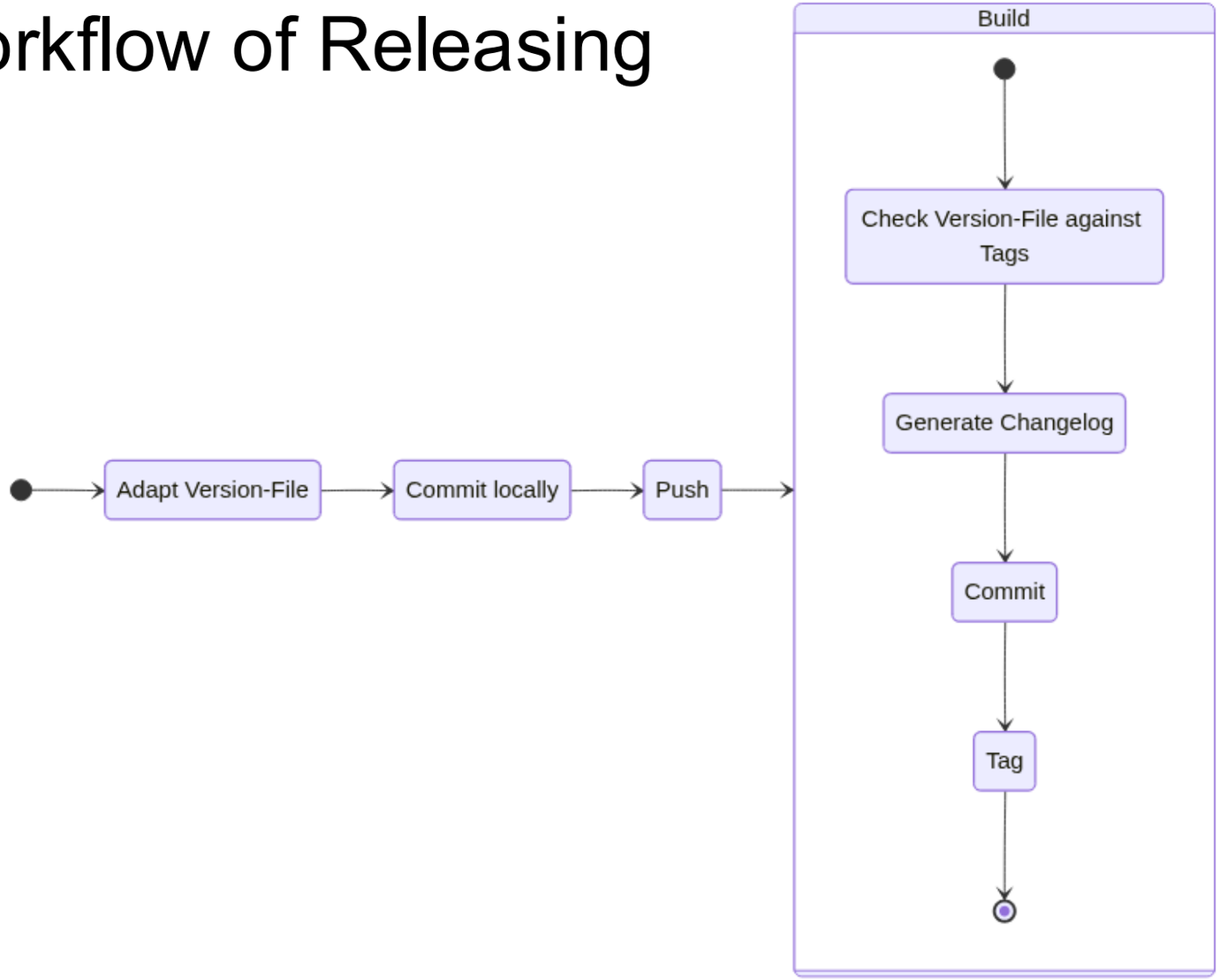


Try to get as close as possible

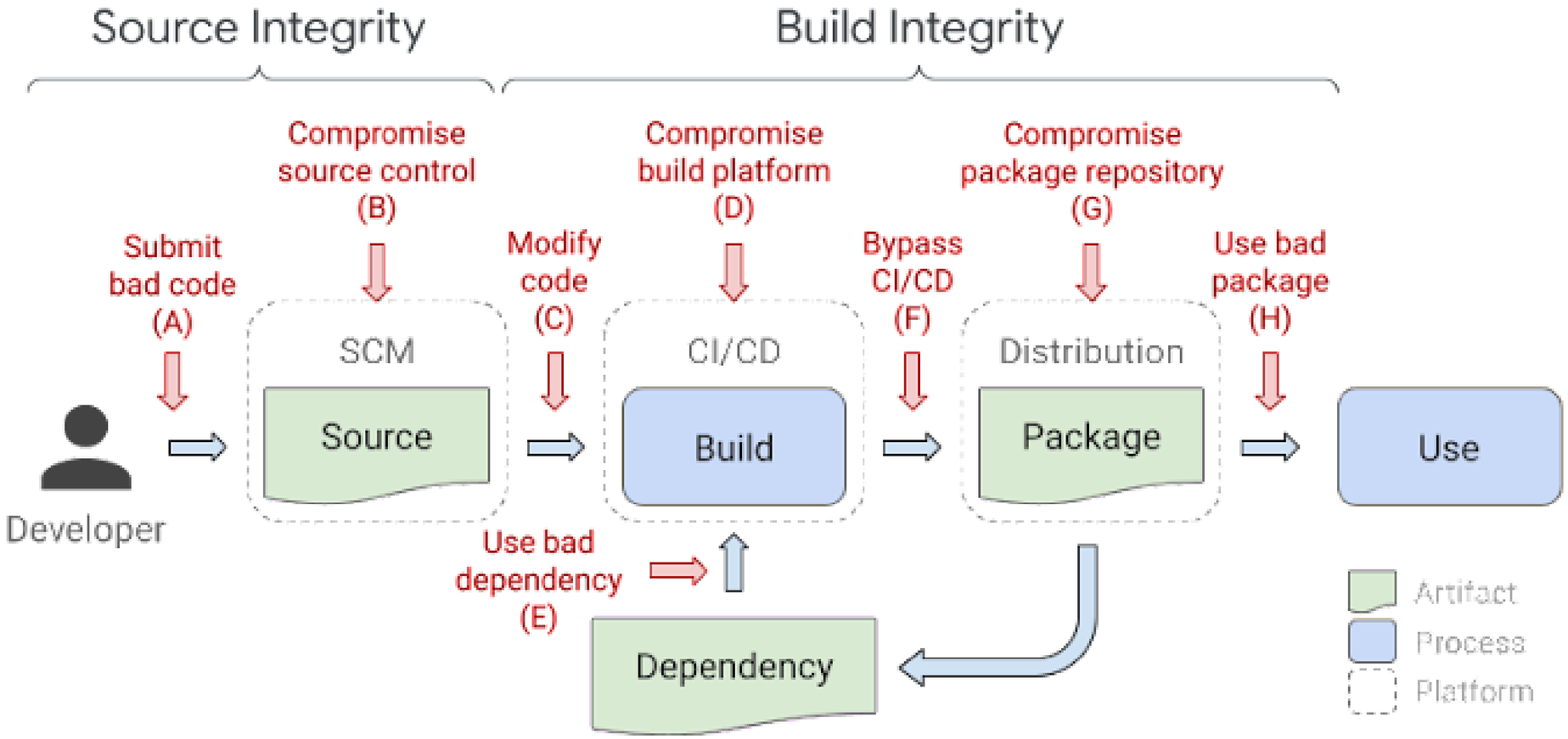
- Sourcecode → Git Tag, Commit
- pom.xml → (Released GAV)
- Build System → (Build Number)
- Docker Image → Tagged Imagename

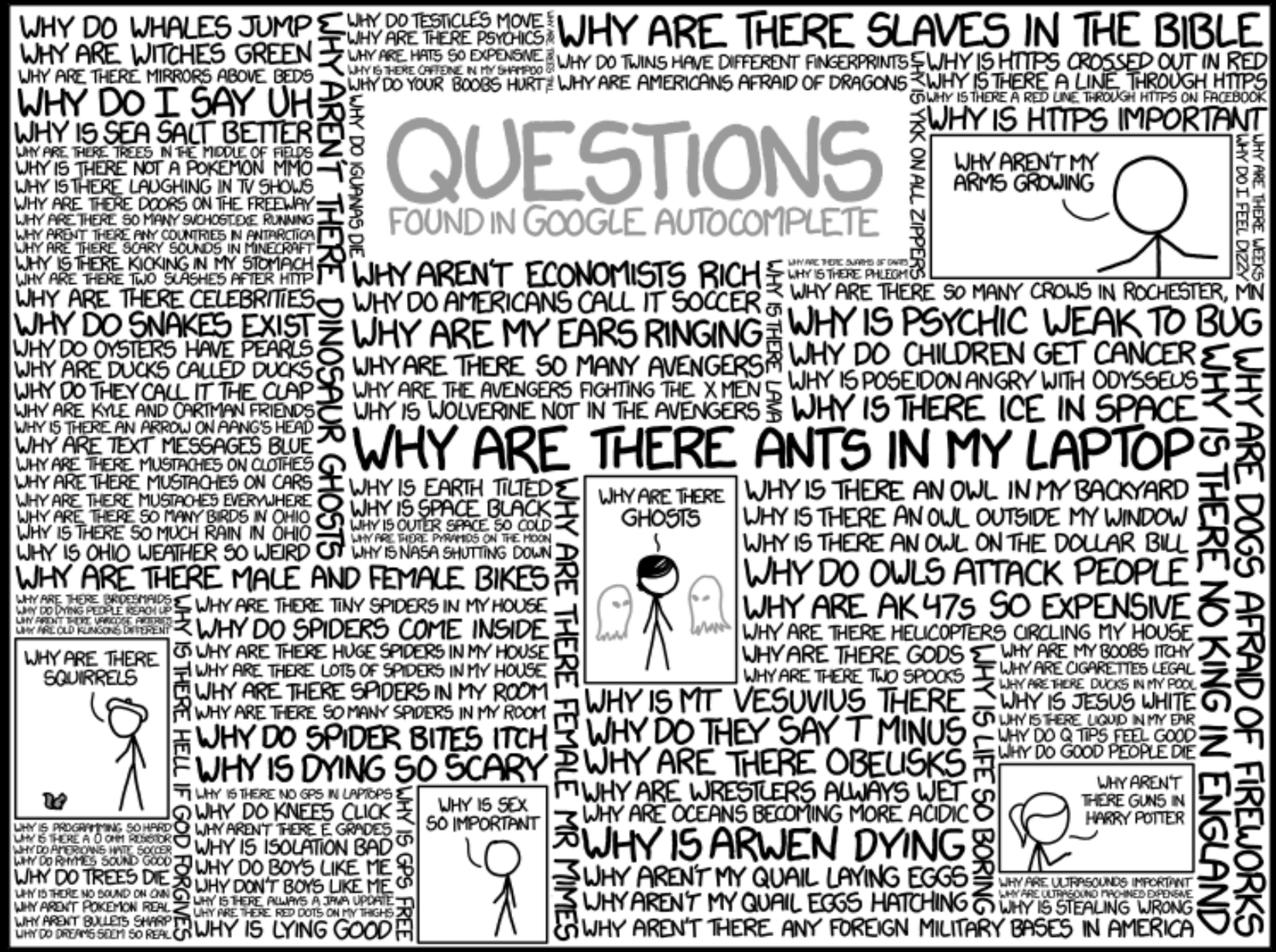
Reference only where applicable,
but always build everything with the same identifiers

Example Workflow of Releasing



Outlook for next session, SLSA




QUESTIONS
FOUND IN GOOGLE AUTOCOMPLETE
WHY DO WHALES JUMP
WHY ARE WITCHES GREEN
WHY ARE THERE MIRRORS ABOVE BEDS
WHY DO I SAY UH
WHY IS SEA SALT BETTER
WHY ARE THERE TREES IN THE MIDDLE OF FIELDS
WHY IS THERE NOT A POKEMON MMO
WHY IS THERE LAUGHING IN TV SHOWS
WHY ARE THERE DOORS ON THE FREEWAY
WHY ARE THERE SO MANY SYCHOSTICKS RUNNING
WHY AREN'T THERE ANY COUNTRIES IN ANTARCTICA
WHY ARE THERE SCARY SOUNDS IN MINECRAFT
WHY IS THERE KICKING IN MY STOMACH
WHY ARE THERE TWO SLASHES AFTER HTTP
WHY ARE THERE CELEBRITIES
WHY DO SNAKES EXIST
WHY DO OYSTERS HAVE PEARLS
WHY ARE DUCKS CALLED DUCKS
WHY DO THEY CALL IT THE CLAP
WHY ARE KYLE AND CARTMAN FRIENDS
WHY IS THERE AN ARROW ON AANG'S HEAD
WHY ARE TEXT MESSAGES BLUE
WHY ARE THERE MUSTACHES ON CLOTHES
WHY ARE THERE MUSTACHES ON CARS
WHY ARE THERE MUSTACHES EVERYWHERE
WHY ARE THERE SO MANY BIRDS IN OHIO
WHY IS THERE SO MUCH RAIN IN OHIO
WHY IS OHIO WEATHER SO WEIRD
WHY ARE THERE MALE AND FEMALE BIKES
WHY ARE THERE BRIDESMAIDS
WHY DO DYING PEOPLE REARM UP
WHY AREN'T THERE VARICOSE AIRBIKES
WHY ARE OLD KLINGONS DIFFERENT
WHY ARE THERE SQUIRRELS
WHY IS PROGRAMMING SO HARD
WHY IS THERE A 0 OHM RESISTOR
WHY DO AMERICANS HATE SOCCER
WHY DO RHYMES SOUND GOOD
WHY DO TREES DIE
WHY IS THERE NO SOUND ON CNN
WHY AREN'T POKEMON REAL
WHY AREN'T BULLETS SHARP
WHY DO DREAMS SEEM SO REAL
WHY ARE THERE TINY SPIDERS IN MY HOUSE
WHY DO SPIDERS COME INSIDE
WHY ARE THERE HUGE SPIDERS IN MY HOUSE
WHY ARE THERE LOTS OF SPIDERS IN MY HOUSE
WHY ARE THERE SPIDERS IN MY ROOM
WHY ARE THERE SO MANY SPIDERS IN MY ROOM
WHY DO SPIDER BITES ITCH
WHY IS DYING SO SCARY
WHY IS THERE NO GPS IN LAPTOPS
WHY DO KNEES CLICK
WHY AREN'T THERE E GRADES
WHY IS ISOLATION BAD
WHY DO BOYS LIKE ME
WHY DON'T BOYS LIKE ME
WHY IS THERE ALWAYS A JAVA UPDATE
WHY ARE THERE RED DOTS ON MY THIGHS
WHY IS LYING GOOD
WHY DO TESTICLES MOVE
WHY ARE THERE PSYCHICS
WHY ARE HATS SO EXPENSIVE
WHY IS THERE CAFFEINE IN MY SHAMPOO
WHY DO YOUR BOOBS HURT
WHY ARE THERE SLAVES IN THE BIBLE
WHY DO TWINS HAVE DIFFERENT FINGERPRINTS
WHY ARE AMERICANS AFRAID OF DRAGONS
WHY IS HTTPS CROSSED OUT IN RED
WHY IS THERE A LINE THROUGH HTTPS
WHY IS THERE A RED LINE THROUGH HTTPS ON FACEBOOK
WHY IS HTTPS IMPORTANT
WHY AREN'T MY ARMS GROWING
WHY IS YKK ON ALL ZIPPERS
WHY ARE THERE WEEKS
WHY DO I FEEL DIZZY
WHY DO IGUANAS DIE
WHY AREN'T ECONOMISTS RICH
WHY DO AMERICANS CALL IT SOCCER
WHY ARE MY EARS RINGING
WHY ARE THERE SO MANY AVENGERS
WHY ARE THE AVENGERS FIGHTING THE X MEN
WHY IS WOLVERINE NOT IN THE AVENGERS
WHY ARE THERE SWARMS OF GNATS
WHY IS THERE PHLEGM
WHY ARE THERE SO MANY CROWS IN ROCHESTER, MN
WHY IS PSYCHIC WEAK TO BUG
WHY DO CHILDREN GET CANCER
WHY IS POSEIDON ANGRY WITH ODYSSEUS
WHY IS THERE ICE IN SPACE
WHY IS THERE LAVA
WHY ARE THERE ANTS IN MY LAPTOP
WHY IS EARTH TILTED
WHY IS SPACE BLACK
WHY IS OUTER SPACE SO COLD
WHY ARE THERE PYRAMIDS ON THE MOON
WHY IS NASA SHUTTING DOWN
WHY ARE THERE GHOSTS
WHY IS THERE AN OWL IN MY BACKYARD
WHY IS THERE AN OWL OUTSIDE MY WINDOW
WHY IS THERE AN OWL ON THE DOLLAR BILL
WHY DO OWLS ATTACK PEOPLE
WHY ARE AK 47s SO EXPENSIVE
WHY ARE THERE HELICOPTERS CIRCLING MY HOUSE
WHY ARE THERE GODS
WHY ARE THERE TWO SPOOKS
WHY IS MT VESUVIUS THERE
WHY DO THEY SAY T MINUS
WHY ARE THERE OBELISKS
WHY ARE WRESTLERS ALWAYS WET
WHY ARE OCEANS BECOMING MORE ACIDIC
WHY IS ARWEN DYING
WHY AREN'T MY QUAIL LAYING EGGS
WHY AREN'T MY QUAIL EGGS HATCHING
WHY AREN'T THERE ANY FOREIGN MILITARY BASES IN AMERICA
WHY IS SEX SO IMPORTANT
WHY ARE MY BOOBS ITCHY
WHY ARE CIGARETTES LEGAL
WHY ARE THERE DUCKS IN MY POOL
WHY IS JESUS WHITE
WHY IS THERE LIQUID IN MY EAR
WHY DO Q TIPS FEEL GOOD
WHY DO GOOD PEOPLE DIE
WHY AREN'T THERE GUNS IN HARRY POTTER
WHY ARE ULTRASOUNDS IMPORTANT
WHY ARE ULTRASOUND MACHINES EXPENSIVE
WHY IS STEALING WRONG
WHY ARE DOGS AFRAID OF FIREWORKS
WHY IS THERE NO KING IN ENGLAND
WHY IS LIFE SO BORING
WHY IS GPS FREE
WHY ARE THERE FEMALE MR MIMES
WHY IS THERE HELL IF GOD FORGIVES
WHY AREN'T THERE DINOSAUR GHOSTS

Recap Questions

1. Why are git tags important for releasing?
2. How should git tags and docker tags be mapped to each other and why?
3. What are reproducible builds?
4. Why are reproducible builds hard to achieve?
5. Why do you need Dagger / Earthly?